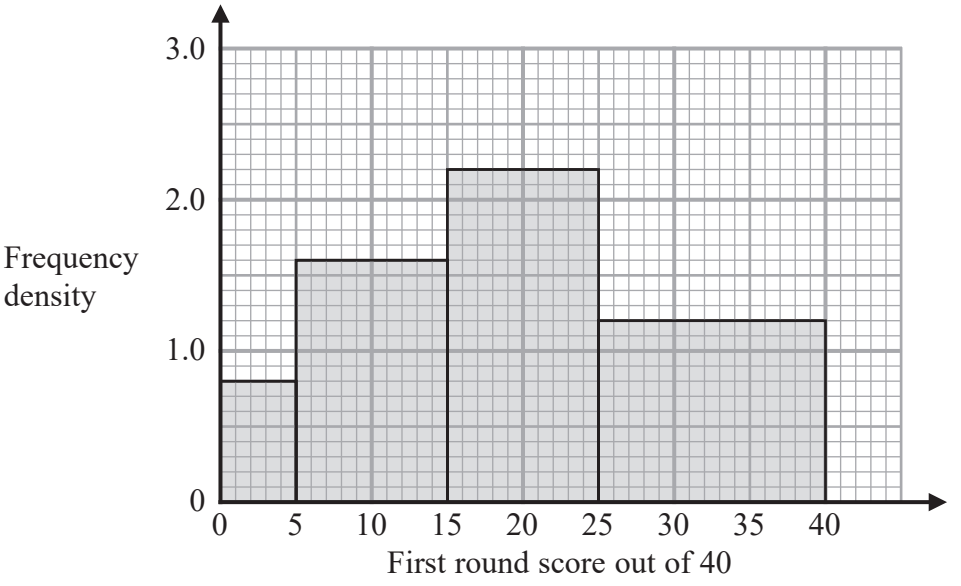
17 Some people took part in the first round of a competition.

The histogram gives information about the scores of these people in the first round.



20% of the people got a score high enough for them to qualify for the second round.

Work out an estimate for the score needed to qualify for the second round.
You must show all your working.

Frequency = class width x frequency density.

0-5: 5(0.8)=4; 5-15: 10(1.6)=16.

15-25: 10(2.2)=22; 25-40: 15(1.2)=18.

Total = 4+16+22+18 = 60 people.

Top 20% = 12 people.

Top class density 1.2, so 12 people span 12/1.2 = 10.

Threshold = 40 - 10 = score 30.

1 mark
one correct class frequency

1 mark
total 60

1 mark
top 20% = 12

1 mark
score 30

30

(Total for Question 17 is 4 marks)

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